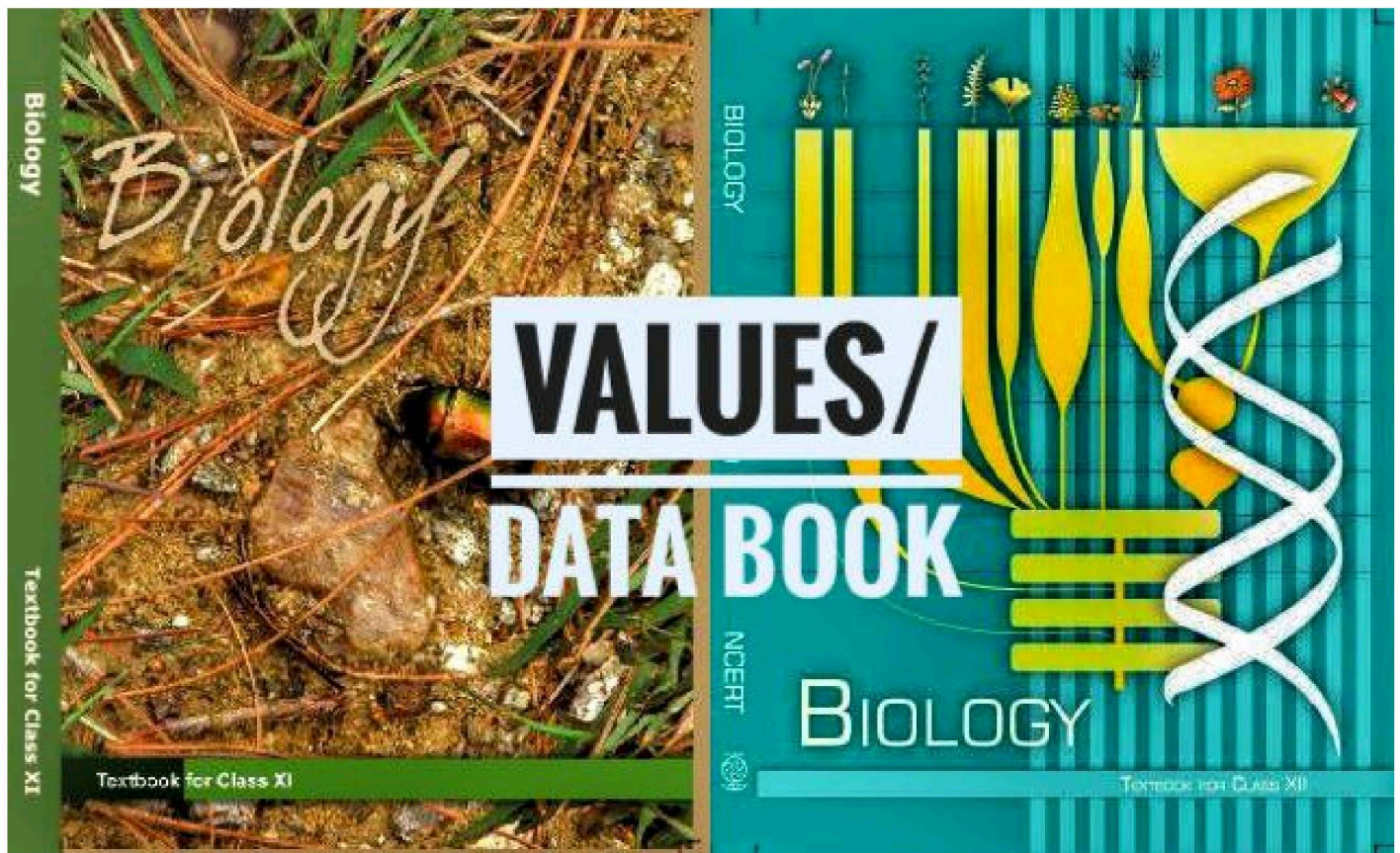


For Quick Revision of NCERT Biology Value based Data

Written by



Numeric Data of NCERT Biology

Diversity

- * Number of known species → 1.7 to 1.8 million (IUCN 2004)
- * R.H. Whittaker proposed five kingdom classification → 1969
- * D.J. Ivanowsky recognise TMV virus which cause mosaic diseases of tobacco. → 1892
- * M.W. Beijerinck demonstrate infectious nature of TMV virus (Contagium vivum fluidum) → 1898
- * T.O. Diener discovered viroid → 1971
- * 50% CO₂ fixation on earth is carried out by algae.
- * 70 species of marine algae used as food.
- * Height of kelps → 100m
- * Taller angiosperm → Eukalyptus (over 100m)
- * $\frac{2}{3}$ (70%) of all species are arthropods.
- * Cyclostomat bear 6-15 pairs gill slits.
- * Bony fishes bear 4 pairs of gill slits.
- * Ctenophora have 8 ciliated comb plate for locomotion
- * more than 50% volume of neural tissue in our body is made up of neuroglial cells.

Lower Animal

- * Earthworm → Clitellum — 14th to 16th segment
 - spermathecal apertures — 5th to 9th segments (ventro-lateral side of intersegmental grooves)
 - Female genital pore — 14th segment (mid ventral)
 - Male genital pore — 18th segment (1 pair) (ventro lateral side)
 - buccal cavity — 1st to 3rd segment
 - oesophagus — 5th - 7th segment

- gizzard → 8th - 9th segment
- Stomach → 9th - 14th segment
- intestine → 15th to last segment
- typhlosole → after 26th segment
- blood glands → 4th, 5th & 6th segment
- septal nephridia → 15th to last (intersegmental septa)
- integumentary nephridia → 3rd to last segment
- pharyngeal nephridia → 4th, 5th, 6th segment (3 pair)
- Nerve cord → 3rd & 4th segment
- Testes → 10th & 11th segment (2 pair)
- Vas deferentia → upto 18th segment
- accessory gland → 17th & 19th segment (2 pair)
- Spermathecae → 6th - 9th segment (4 pair)
- Ovary → 12th & 13th segment (1 pair)
(inter segmental septum)
- 1 Cocoon produce → 2-20 baby worm
(after 3 week)

Cockroach

- size ⇒ $\frac{1}{4}$ to 3 inch (0.6 - 7.6 cm)
- *Periplaneta americana* ⇒ 34 - 53 mm long
- triangular head formed by fusion of 6 segments
- Abdomen → 10 segmented
- brood pouch → 7th, 8th & 9th sternum
- genital pouch → 9th, 10th terga & 9th sternum
- anal cerci → 10th segment
- 1 eye bears 2000 hexagonal ommatidia.
- hepatic caeca → 6-8 blind tubule
- Malpighian tubule → 100-150 (yellow colour)
- 10 pair of spiracles → lateral side of body.

- if head cut off $\rightarrow$ still live for 1 week.
- 1 pair testes $\rightarrow$ 4th - 5th abdominal segment (lateral side)
- Mushroom glands $\rightarrow$ 6th - 7th abdominal segment
- 2 ovaries $\rightarrow$ 2 - 6 abdominal segment
- each ovary formed by 8 ovarian tubule/ovarioles
- 1 pair spermatheca $\rightarrow$ 6th abdominal segment
- size of oothecae $\rightarrow \frac{3}{8}$ " (8 mm)
- 1 ootheca contain $\rightarrow$ 14 to 16 egg
- female produce $\rightarrow$ 9 to 10 ootheca
- Nymph grow by moulting about 13 times to reach adult form

Frog

- * In frog 10-12 vasa efferentia present in each testis.
- * A mature female frog lay 2500 to 3000 ova at a time

Cell

- * Discovery of nucleus by Robert Brown $\rightarrow$ 1831
- * Cell theory $\rightarrow$ Matthias Schleiden (1838) - German
Theodore Schwann (1839) - British
Rudolf Virchow (1855)
- * Mycoplasma $\rightarrow$ 0.3 μ m
PPLO $\rightarrow$ 0.1 μ m
- Bacteria $\rightarrow$ 3 to 5 μ m
- Typical bacteria $\rightarrow$ 1-2 μ m
- RBC $\rightarrow$ 7 μ m in diameter
- viruses $\rightarrow$ 0.02 μ m to 0.2 μ m
- * Ribosome $\rightarrow$ 15 X 20 nm
- * Membrane of RBC $\Rightarrow$ 52% protein &
40% lipid
- * Invention of e- microscope $\rightarrow$ 1950s

- * Fluid Mosaic Model of plasma membrane by Singer & Nicolson → 1972
- * Golgi body → discovered by Camillo Golgi → 1898
↳ flat disc shaped cisternal → 0.5-1 μm in diameter
- * 90% volume of plant cell occupied by vacuoles.
- * Mitochondria
 - ↳ Diameter → 0.2-1 μm (average 0.5 μm)
 - ↳ length → 1-4.1 μm
- * Chloroplast
 - ↳ length → 5-10 μm
 - ↳ width → 2-4 μm
- * Number of chloroplast in
 - Chlamydomonas = 1
 - Mesophyll cell = 20-40
- * Discovery of Ribosome → (1953) by George Palade
- * Cilia & Flagella → filament → 9+2 } microtubule arrangement
 basal body → 9+0 }
- * Centriole → 9+0 microtubule arrangement.
- * Perinuclear space → 10-50 nm
- Biomolecules
- * R-group of fatty acid ⇒ 1 to 19 Carbon
- * Arachidonic acid have 20 Carbon including R-group.
- * Palmitic acid have 16 - - - - -
- * Biomolecules
 - ↳ micromolecule → mw < 1,000 Dalton
 - ↳ macromolecule → mw ≥ 10,000 Dalton

Lipids → mw ≤ 800 dalton
- * There are 20 types of amino acids
 - ↳ 10 essential
 - ↳ 10 non essential

* Average composition of cell

Water $\rightarrow$ 70-90 %
Protein $\rightarrow$ 10-15 %
Carbohydrate $\rightarrow$ 3 %
Lipids $\rightarrow$ 2 %
Nucleic Acids $\rightarrow$ 5-7 %
Ions $\rightarrow$ 1 %

* Pitch/Length of Helix of DNA $\Rightarrow$ 34 Å (B-DNA)

* Glycolysis $\rightarrow$ 10 metabolic steps with 10 enzymes.

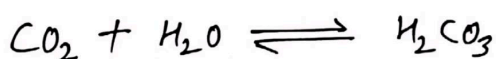
* Blood glucose level in a healthy person $\rightarrow$ 80-100 mg/dl
 $\Rightarrow$ 4.5-5 mM

* enzymes get damaged above 40°C temperature

* enzymes from eurythermal organism are functional upto 80-90°C temperature.

* Thumb rule $\Rightarrow$ rate of rxn $\left\{ \begin{array}{l} \text{doubled} \\ \text{decreased} \end{array} \right.$ for every 10°C temperature change by half

* Rxn



$r = 200 \text{ molecule/hr}$ (without enzyme)

$r = 6 \text{ lakh molecule/sec}$ (with Carbonic anhydrase enzyme)

* Enzymes divided into 6 Class & each class further divided into 4-13 subclasses. & named by a 4 digit number.

Cell Cycle

* Human cell divide once in every 24 hr.

* Cell cycle of yeast $\rightarrow$ 90 minutes.

* M-phase of human cell cycle $\rightarrow$ 1 hr

$\hookrightarrow$ Interphase $\rightarrow$ more than 95 %

* Cell cycle of *E. coli* = 20 minutes

PLANT WATER RELATION

* Types of aquaporins = 8 (water channel)

* Watermelon have more than 92% water.

* Dry matter of herbaceous plants $\Rightarrow$ 10-15% of fresh weight

* A dry seed still have 10-15% water of its weight.

* A corn plant absorb 3 liter water in a day.

* A mustard plant absorb water equal to its weight in about 5 hr.

* By diffusion $\rightarrow$ movement of molecule across a typical plant cell (50 μm) takes approximately 2.5 s.

* Speed of water upward in xylem by transpiration pull $\rightarrow$ upto 15 m/hr

* Less than 1% of water reaching the leaves is used in photosynthesis & plant growth. 98% of it is lost by transpiration.

* Transpiration can create pressure sufficient to lift a xylem sized column of water over 130 m high.

* Transpiration cools leaf sometimes 10-15°C by evaporative cooling.

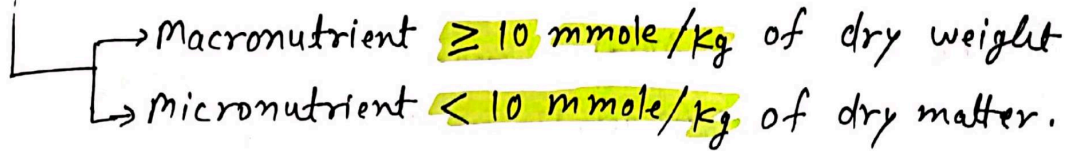
MINERAL NUTRITION

* Hydroponics technique $\rightarrow$ Julius von Sachs (1860)

$\hookrightarrow$ able to detect very low (10^{-8}g/l) concentration of mineral present in plants

* More than 60 ~~ene~~ elements are found in different plants.

* Concentration of nutrient in plant



* There are 17 essential element & 4 beneficial element for plants.
└─> macro & micro nutrient
└─> Na, Si, Co, Se

PHOTOSYNTHESIS

* During N_2 fixation nitrogenase enzyme require 8 ATP for 1 NH_3 production.

* Discovery of oxygen $\rightarrow$ by Joseph Priestley (1774)

* Priestley's experiment $\Rightarrow$ 1770 (Bell jar, mint plant, mice, candle)

* 1854 $\rightarrow$ Julius von Sachs prove that glucose is formed in photosynthesis & stored as starch.

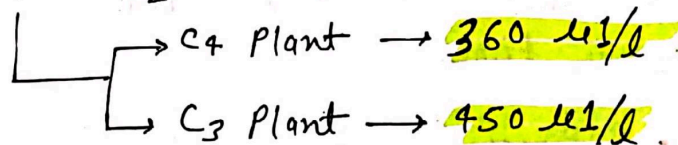
* 1 Kelvin Cycle (C_3 Cycle) $\rightarrow$ intake of 1 CO_2 , 3 ATP & 2 $NADPH_2$

└─> 6 cycle run for 1 glucose formation

* Law of limiting factor $\rightarrow$ (1905) by Blackman

* Light saturation occurs at 10% of full sunlight.

* Saturation for CO_2 concentration



* CO_2 concentration in atmosphere $\rightarrow$ 0.03 - 0.04 %

Respiration in Plants

* In Fermentation more than 13% alcohol concentration is poisonous for yeast itself.

→ so distillation is required

* Respiration → Aerobic respiration → 36/38 ATP form
→ Anaerobic respiration → 2 ATP formation
(Glycolysis) (less than 7% of aerobic respⁿ)

* Breakdown of 1 glucose gives

Glycolysis → <u>2</u>	→ <u>6</u> CO ₂	→ <u>2</u> Link Rx ⁿ
Link Rx ⁿ → <u>2</u>		→ <u>4</u> Krebs Cycle
Krebs Cycle → <u>6</u>		→ <u>10</u> NADH ₂ → <u>30</u> ATP
	Krebs Cycle →	→ <u>2</u> FADH ₂ → <u>4</u> ATP
	Glycolysis →	→ <u>2</u> ATP → <u>2</u> ATP
	Krebs Cycle →	→ <u>2</u> GTP → <u>2</u> ATP
		Total → <u>38</u> ATP

* Respiratory Quotient → 1 for Carbohydrate
→ less than 1 for fats
→ 0.7 for Tripalmitin (fatty acid)
→ 0.9 for proteins.

Plant Growth

* Root apical meristem give rise to more than 17500 new cells per hour.

* Cells in watermelon may increase size by up to 3,50,000 times.

* During mid 1960s purification & chemical characterisation of 3 different type of inhibitors (ABA)

* There are more than 100 gibberellins reported from different organism such as fungi & higher plants.

* Spraying sugarcane crop with gibberellins increase the length of stem, thus increasing the yield as much as 20 tonnes per acre.

Digestion

* Dental formula — [Child → $\frac{2102}{2102}$ (Premolar absent)
Adult → $\frac{2123}{2123}$ (But no any diastema)
 $\frac{212}{212}$

* Weight of liver → 1.2 – 1.5 Kg

* pH — [Saliva → 6.8
Gastric juice → 1.8
Intestine → 7.8

* Stomach store food for 4-5 hour.

* 30% Starch digested by salivary amylase & 70% is digested by pancreatic amylase.

(in Kcal/g)	Carbohydrate	Protein	Fat
Gross calorific Value → 4.1	5.65	9.45	
Physiological Calorific Value → 4.0	4.0	9.0	

Respiration

* Breathing rate → 12 – 16 times/minute

* Respiratory volume & capacities

(TV) Tidal Volume ⇒ 500 ml
IRV ⇒ 2500 – 3000 ml
ERV ⇒ 1000 – 1100 ml
(RV) Residual Volume ⇒ 1100 – 1200 ml
IC = TV + IRV ⇒ 3000 – 3500 ml
EC = TV + ERV ⇒ 1500 – 1600 ml
FRC = ERV + RV ⇒ 2100 – 2300 ml
ERV + IRV + TV = VC ⇒ 4000 – 4600 ml
TLC = VC + RV ⇒ 5100 – 5800 ml

* $pO_2 \Rightarrow$ (in mmHg)	<u>Air</u> 159	Alveoli 104	Oxygenated Blood 95	Deoxy blood 40	Tissue 40
$pCO_2 \Rightarrow$	0.3	40	40	45	45

* Solubility of CO_2 is 20-25 times higher than that of O_2 .

* Thickness of diffusion membrane of alveoli = 0.2 mm

* Solubility of CO is 300 times higher than O_2 in blood.

* Transport of gases

$O_2 \Rightarrow$ 97% as oxy Hb & 3% dissolve in plasma
in RBC

$CO_2 \Rightarrow$ 20-25% as Carbamino Hb & 70% as
Bicarbonate in RBC & 7% in plasma.

* Each haemoglobin molecule can carry maximum 4 molecule of O_2 .

* Every 100 ml of oxygenated blood can deliver around 5ml of O_2 to the tissue under normal physiological condition.

* Every 100 ml of deoxygenated blood deliver approximately 4ml of CO_2 to alveoli.

Circulatory

* Blood $\rightarrow$ 55% Plasma $\rightarrow$ 90-92% water
6-8% protein
45% formed element $\rightarrow$ RBC, WBC, Platelets

* RBC $\rightarrow$ 5-5.5 million/mm³ of blood.

$\rightarrow$ life span $\rightarrow$ 120 days

* WBC $\rightarrow$ 6000-8000/mm³ of blood

* WBC — { Basophil → 0.5-1%
Eosinophil → 2-3%
Monocyte → 6-8%
Lymphocyte → 20-25%
Neutrophils → 60-65% } of total WBC

* Platelets/Thrombocytes → 1.5-3.5 lakh/mm³ of blood.

* Number of chamber of Heart — { 2 → Fishes
3 → Amphibia, Reptile
4 → Birds, mammals, crocodile

* Number of impulse (Action potential) By — { SAN → 70-75/min
↓
AVN → Heart beat (72/min average)

* 70% blood passively flow from atrium to ventricle & 30% blood flow by atrial systole.

* Duration of a cardiac cycle is 0.8 second.

* Each ventricle pumps out approximately 70 ml of blood ⇒ stroke volume

* Cardiac output = Heart beat × Stroke volume
= 72 × 70
= 5000 ml/min

* Normal blood pressure ⇒ 120/80 mmHg

* Hypertension ⇒ 140/90 mmHg @ higher.

Excretion

* Kidney — { length → 10-12 cm
width → 5-7 cm
thickness → 2-3 cm
weight → 120-170 g

* Each kidney have 1 million nephron.

- * Renal blood flow $\Rightarrow 1100 - 1200 \text{ ml/min}$
 $\downarrow$
 $\frac{1}{5}$ of cardiac output
- * G.F.R. $= 125 \text{ ml/min} = 180 \text{ liter/day}$
 its 99% is reabsorbed
- * Urine released $1-1.5 \text{ liter/day}$.
- * $70-80\%$ reabsorption occur in PCT .
 $\downarrow$
 (electrolyte + water) maximum reabsorption
- * Osmolarity of kidney $\left\{ \begin{array}{l} \text{Cortex} \rightarrow 300 \text{ mOsmol/liter} \\ \text{medulla} \rightarrow 1200 \text{ } \end{array} \right.$
- * Osmolarity of blood $\Rightarrow 300 \text{ mOsmol/liter}$
- * Human kidney can produce 4 times concentrated urine than initial filtrate formed.
- * $25-30 \text{ g}$ of urea is excreted out per day.
- * Normal blood urea level $\Rightarrow 17-30 \text{ mg/100ml}$
- * Our lungs remove $\text{CO}_2 \rightarrow 200 \text{ ml/min}$.

Locomotion

- * $40-50\%$ of body weight is by muscles.
- * Skeletal system $\Rightarrow 206 \text{ bone}$
 - $\rightarrow$ Axial $\rightarrow 80$ (Skull (29) + vertebral column (26) + Sternum (17) + Ribs (24))
 - $\rightarrow$ Appendicular $\rightarrow$ Limb (30 x 4) (126) + Pectoral girdle (4) + Pelvic girdle (2)
- * Skull (29) $\left\{ \begin{array}{l} \text{Cranium} - 8 \\ \text{Face} - 14 \\ \text{ear ossicle} - 6 \\ \text{Hyoid} - 1 \end{array} \right\} 22$

* Vertebral column $\rightarrow C_7 T_{12} L_5 S_1 C_1$

* Ribs (12 pair) $\left\{ \begin{array}{l} \text{Vertebral} \rightarrow 1^{st} - 7^{th} \text{ pair} \\ \text{vertebrochondral} \rightarrow 8^{th}, 9^{th}, 10^{th} \text{ pair} \\ \text{floating} \rightarrow 11^{th} \& 12^{th} \text{ pair} \end{array} \right.$

* Nervous System

* $Na^+ - K^+$ pump transport $3 Na^+$ outward for $2 K^+$ into the cell.

* Resting membrane potential $\Rightarrow -70 mV$

* speed of nerve conduction in mammals $\Rightarrow 100 - 130 m/s$

Endocrine

* $1 \text{ to } 2 \%$ of pancreas have $1 - 2$ million Islets of Langerhans.

Reproduction in Organism

* Life Span $\left\{ \begin{array}{l} \text{Butterfly} \rightarrow 1 - 2 \text{ week} \\ \text{Crow} \rightarrow 15 \text{ years} \\ \text{Crocodile} \rightarrow 50 \text{ years} \\ \text{Tortoise} \rightarrow 100 - 150 \text{ years} \end{array} \right.$

* Cyst of Amoeba $\Rightarrow$ 3 layered hard covering

* Bamboo flower only once after $50 - 100$ years.

* *Strobilanthes kunthiana* flower once in 12 years.

* Number of Chromosome in meiocyte (2n) September - October

Human - 46

House fly - 12

Rat - 42

Dog - 78

Cat - 38

Fruit fly - 8

Ophioglossum (a fern) - 1260

Apple - 34

Rice - 24

Maze - 20

Potato - 48

Butterfly - 380

Onion - 16

Honey bee - 32
(Queen/Worker)

Drone - 16

(n)

Sexual reproduction in flowering plant

- * Pollen grain are generally spherical $\rightarrow$ 20-25 μm in diameter.
- * 60% angiosperm pollen grain shed in 2 celled stage.
- * 40% angiosperm pollen grain shed in 3 celled stage.
- * Pollen grain viability of $\rightarrow$ rice & wheat $\rightarrow$ 30 min.

$\hookrightarrow$ Rosaceae, Leguminosae & Solanaceae $\rightarrow$ 1 month

- * Mature female gametophyte of angiosperm = Embryo sac

$\downarrow$
7-celled, 8-nucleated

- * Pollination by water is rare in angiosperm and is limited to about 30 genera, mostly monocot.
- * Tallest flower $\rightarrow$ Amorphophallus $\rightarrow$ 6 feet in height.
- * For pollen germination we can use 10% sugar solution. After 15-30 min we can see pollen tube with low power of microscope.
- * The oldest viable seed is that of a Lupine (*Lupinus arcticus*) excavated from Arctic Tundra. It germinated & flowered after 10,000 years of dormancy.
- * Date palm (*Phoenix dactylifera*) discovered during archeological excavation at King Herod's palace near dead sea $\Rightarrow$ 2000 year old viable seed.

Human Reproduction

- * Scrotum helps in maintaining the low temperature of testes (2-2.5°C lower than normal body temp)
- * Testes $\rightarrow$ length $\rightarrow$ 4-5 cm
 $\rightarrow$ width $\rightarrow$ 2-3 cm
 $\nearrow$ contain 1 to 3 Seminiferous tubule
- * Each testis has about 250 testicular lobules.
- * Length of each ovary $\rightarrow$ 2-4 cm
- * Length of fallopian tube $\rightarrow$ 10-12 cm

- * The glandular tissue of each breast is divided into **15-20** mammary lobe.
- * Human male ejaculate about **200-300 million** sperms during a coitus.
- * For normal fertility **60%** of sperms have normal shape & size and for at least **40%** of them must show vigorous motility.
- * During the embryonic development stage **2 million** oogonia are formed within each fetal ovary. ^{million}
- * At puberty only **60,000-80,000** primary follicle are left in each ovary. $\Rightarrow (120000 - 160000)$
- * Menstrual cycle (**28** days) $\left\{ \begin{array}{l} \text{menstrual flow} \rightarrow \text{3-5 days} \\ \text{ovulation} \rightarrow \text{14th day} \end{array} \right.$
- * In Human menstruation (menstrual cycle) ceases around **50 years** of age $\Rightarrow$ Menopause.
- * Embryo with **8-16** blastomere is called morula. (Solid Ball)
- * The human pregnancy lasts **9 months**.
- * During pregnancy $\rightarrow$ **1st** month $\rightarrow$ Heart formed
 - $\rightarrow$ By the end of **2nd** month $\rightarrow$ Limb & digits formed.
 - $\rightarrow$ By the end of **12 week** (**1st trimester**) @ **3rd** month $\rightarrow$ most of major organs formed like limbs & external genitals are well developed.
 - $\rightarrow$ During **5th** month $\rightarrow$ **1st** movement of foetus & Hair on head observed.
 - $\rightarrow$ By the end of **24 weeks** (**2nd trimester** @ **6th** month) $\rightarrow$ Body covered with fine hairs, eyelid seperated & eye lashes are formed.
 - $\rightarrow$ By the end of **9 month** $\rightarrow$ foetus is fully developed.

Reproductive Health

- * Family planning were initiated in 1951.
- * In 1900 — world population → 2 billion
 ↳ India population → 350 million (in 1947)
- * In 2000 — world population → 6 billion
 ↳ India population → 1 billion (crossed in May)
- * According to 2001 census report, the population growth rate was still around 1.7% i.e. → 17/1000/year.
 by this rate our population could double in 33 years.
- * Periodic abstinence → 10th–17th day of menstrual cycle
 couple avoid from coitus to prevent unwanted pregnancy.
 → (5th to 26th day)
- * Oral contraceptive pills — taken daily (21 day continuous)
 ↳ after gap of 7 days (during menstruation)
- * Saheli is non steriodal 1 week pill.
- * Emergency pills & IUDs within 72 hour of coitus have been found to be very effective.
- * Nearly 45–50 million MTPs are performed in a year all over the world which is 1/5th of total conceived pregnancy in a year.
- * Govt. of India legalised MTP in 1971.
- * MTPs are considered relatively safe during the 1st trimester i.e. upto 12 week/3 month of pregnancy.
- * The age group of 15–24 years persons are very high vulnerable to STDs.
- * Inability to conceive even after 2 years of unprotected coitus is called infertility.

Principles of Inheritance

- * Mendel work on garden pea for 7 years (1856-1863)
- * Mendel select 14 true breeding pea plant varieties.
- * Mendel publish his work on inheritance of character in 1865 but it remain unrecognised till 1900.
- * 1900 → Rediscovery of Mendel's work by Hugo de Vries, Carl Correns & Von Tschermak
- * 1902 → Chromosomal theory of inheritance given by Sutton and Boveri.
- * Morgan's experiment on *Drosophila*
 - Body colour & eye colour genes are tightly linked & show only 1.3% recombination.
 - Eye colour & Body size gene show 37.2% recombination
- * Henking (1891) - discovered X-body (X-chromosome) & XO type of sex determination in insect.
- * Colour blindness occurs in about 8% of males & only about 0.4% of females in the world.
- * In sickle cell anemia point mutation occurs in DNA & 6th codon of m-RNA of β -globulin chain → GAG to GUG
- * Down's Syndrome → Trisomy of 21st chromosome
 - ↳ discovered by Langdon Down (1866)
- * Thalassemia
 - α - 2 gene from each parent = 4 gene (HBA1, HBA2) → Chromosome - 16th
 - β - 1 gene from each parent = 2 gene (HBB) → Chromosome → 11th
 - ↓ mutation in gene
 - ↓ Quantitative disorder

Molecular Basis of Inheritance

- * Bacteriophage $\phi 174$ has 5386 nucleotides.
- * Bacteriophage lambda has 48502 base pair.
- * Escherichia coli has 4.6×10^6 base pair (bp)
- * Human DNA has 6.6×10^9 bp in diploid cell.
- * DNA was first identified by Friedrich Meischer in 1869. He named it as Nuclein.
- * Distance between two base pair in DNA is 3.4 Å & total number of base pair is 6.6×10^9 . So length of DNA is approx 2.2 m.
- * A typical nucleosome contain 200 bp of DNA.
- * 1928 → Frederick Griffith experiment with Streptococcus pneumoniae bacteria ⇒ Transformation.
- * 1933-44 → Oswald Avery, Colin MacLeod and Maclyn McCarty use transforming principle of Griffith's experiment and concluded that DNA is genetic material.
- * 1952 → Alfred Hershey & Martha Chase experiment with bacteriophage, E. coli bacteria & radioactive isotopes of ^{35}S & ^{32}P ⇒ 3 step experiment
Prove DNA is genetic material ←
 - infection
 - Blending
 - Centrifugation
- * 1958 → Matthew Meselson & Franklin Stahl experimentally prove the semi conservative nature of DNA replication. → grow E. coli on $^{15}\text{NH}_4\text{Cl}$ medium.
- * E. coli divide in every 20 minute but complete its DNA replication in 38 minutes.
↳ division start होने से पहले Replication start हो जाता है।

- * 50% of discovered gene's functions are unknown.
- * less than 2% of genome code for protein.
- * Chromosome 1 has most genes (2968) and Y has the fewest (231).
- * Scientists identified 1.4 million SNPs in Human.
- * less than 2% of genome code for protein
i.e. 98% of genome do not code.
 - ↳ high degree of Polymorphism (RFLP)
 - ↳ Basis of DNA Fingerprinting
- * If frequency of an allele in population is greater than 0.01%, it shows that the new allele formed by mutation which regulate different character now.
- * Size of VNTR $\rightarrow$ 0.1 to 20 kb.

Evolution

- * The universe is almost 20 billion years old.
- * Earth was supposed to have been formed about 4.5 billion year back.
- * Life appeared 500 million years after the formation of earth, i.e. 4 billion years back. (≈ 3.9 bya)
- * 1953 - S.L. Miller experiment
 - ↳ with water vapour
 - ↳ $\text{CH}_4 : \text{NH}_3 : \text{H}_2 \Rightarrow 2 : 1 : 2$
 - ↳ 800°C
- * First non-cellular form of life could have originated 3 billion year back.
- * The first cellular form of life did not possibly originated till 2000 million year ago. (2 bya)
- * According to conventional religious literature
 - ↳ Theory of special creation
 - ↳ earth is about 4000 years old.

- * Triceratops → having 3 horn
- * Tyrannosaurus → 20 feet long.
- * 1850s → Before industrialisation → more white wing moth.
- * In 1920 → after industrialisation → more dark wing moth.
- * About 2 billion years ago first cellular form (Prokaryot) of life appeared on earth.
- * Invertebrates were formed & active → 500 mya
- * Jawless fishes evolved around → 350 mya
- * See weeds & few plants exist around → 320 mya
- * Fish with stout & strong fins could move on land and go back to water → 350 mya
- * 1938 → Coelacanth/lobefines Caught in South Africa.
 - ↳ Ancestors of amphibia
- * Reptiles of different shape and size dominate on earth in the next 200 million years (200 mya)
- * Fish like reptile evolved → 200 mya
 - ↳ Ichthyosaurs
- * Dinosaurs suddenly disappear from earth about 65 mya.
- * Primates called Dryopithecus and Ramapithecus were existing about 15 mya.
- * About 3-4 mya man like primates walked in eastern Africa. They are not taller than 4 feet.
- * Australopithecus lived in East African grassland about 2 mya.
- * Cranial capacity of Homo habilis → 650-800 cc
- * Java man (Homo erectus) lived → 1.5 mya
 - ↳ Cranial capacity → 900 cc
- * Fossils of Java man discovered in Java in 1891
- * Neanderthal man lived in near east & central Asia between 1 lakh to 40,000 years back.
- * Cranial capacity of Neanderthal man → 1400 cc.

- * During ice age $\rightarrow$ 75000 - 10000 years ago modern *Homo sapiens* arose.
- * Pre historic cave art developed about 18000 years ago.
- * Agriculture came around 10000 years back and human settlement started.

* Human Health and Diseases

- * In typhoid fever body temperature $\rightarrow$ 39° to 40°C
- * Common cold usually last for 3-7 days.
- * No. of disulphide bond present in an antibody molecule is $\rightarrow$ 16 H_2L_2
- * Mucosal-associated lymphoid tissue (MALT) constitutes about 50% of lymphoid tissue in human body.
- * AIDS was first reported in 1981.
- * In the last 25 years or so AIDS spread all over the world killing more than 25 million person.
- * Incubation period of AIDS $\rightarrow$ 5 to 10 years.
- * More than 1 million people suffer from cancer in India and a large number of them die from it annually.
- * Tobacco has been used by human being for more than 400 years.
- * Adolescence period $\rightarrow$ 12-18 years of age.

* Strategies for Enhancement in Food Production

- * 70% of the world livestock population is in India and China but its contribution to the world farm produce is only 25%.
- * Inbreeding $\rightarrow$ mating between closely related individual within the same breed for 4-6 generation.
- * Out crossing $\rightarrow$ mating between animal of same breed but having no common ancestors on either side of their pedigree up to 4-6 generation.

- * In MOET by FSH induced super ovulation
i.e. - 6-8 egg per cycle.
- * The fertilised eggs at 8-32 cells stages are recovered non surgically & transferred to surrogate mother.
- * Conventional plant breeding starts → 9000-11000 years ago. → usually only 1 in few 100 to 1000 cross show desirable combination.
- * In Plant breeding, Evaluation of new variety of plant in research field is followed by testing the materials in farmers' field, for at least 3 growing seasons at several locations in the country.
- * Agriculture accounts for approximately 33% of India's GDP and employs nearly 62% of the population.
- * Green Revolution → mid 1960s → Tripling the food supply.
- * During Period 1960-2000
 - Wheat production ↑ ⇒ 11 million tonnes to 75 million tonnes.
 - Rice production ↑ ⇒ 35 million tonnes to 89.5 million tonnes.
- * In 1963 several varieties of wheat introduced in India. Sonalika → high yielding & Kalyan Sona → diseases resistant
- * In 1966 derivatives of semidwarf variety of rice introduced in India. later better yielding semi dwarf varieties Jaya and Ratna were developed in India.
- * Crop losses can often upto 20-30% (or) even 100% for some times. suffering from hidden hunger
- * More than 840 million people do not have adequate food. more than 3 billion people
- * In 2000, maize hybrid developed in which amount of amino acid is doubled compare to existing maize variety.
- * It takes 3-10 kg of grain to produce 1 kg of meat by animal farming.

- * More than 25% of human population is suffering from hunger and malnutrition.
- * It has been calculated that a 250 kg cow produce 200 g of protein per day.
- * A microorganism like *Methylophilus methylotrophus* can be expected to produce 25 tonnes of protein by 250 g of organism in 1 day.
- * Tissue culture → 1950, → whole plant could be generated from explant.

* Microbes in Human Welfare

- * Fleming, Chain and Florey were awarded the Noble Prize of 1945 for discovery of Penicillin.
- * Biogas plant → 10-15 feet deep tank.

Biotechnology: Principles and Process

- * In 1972 Stanley Cohen and Herbert Boyer created 1st recombinant DNA by introducing antibiotic resistant gene in plasmid of *Salmonella typhimurium*.
- * In 1963 restriction endonuclease enzyme were isolated from *E. coli* which restrict growth of bacteriophage.
- * Restriction endonuclease recognize a specific sequence of 6 base pair. ⇒ palindromic nucleotide sequence.
- * Today we know more than 900 restriction enzymes that have been isolated from 230 strains of bacteria.
- * To make competent host for transformation with recombinant gene ⇒ by incubating the host cell with rDNA on ice, followed by heat shock (42°C), and then putting them back on ice.

* PCR

→ Denaturation at 94°C
→ Annealing at 54°C
→ Extension at 72°C

* 30 cycle of PCR $\Rightarrow 2^{30} \approx 1$ billion copy of DNA.

* Volume of bioreactors = 100-1000 liters

✱

Biotechnology and Its Applications

* 3 critical research areas of biotechnology

→ providing best catalyst
→ creating optimal condition
→ Downstream processing

* At present about 30 recombinant therapeutics have been approved for human use the world over.

* In India 12 of these therapeutics are presently being marketed.

* In 1983 Eli Lilly American company prepare human insulin from E. coli by introducing DNA sequence of A & B peptide chain into the plasmid of E. coli.

* first clinical gene therapy was given in 1990 to a 4 year old girl with ADA deficiency.

* 95% of all existing transgenic animals are mice.

* In 1997, first transgenic cow, Rosie, produced human protein enriched milk (2.4 gm/liter)
→ α -Lactalbumin.

* There are an estimated 2 lakh varieties of rice in India alone.

* 27 documented varieties of Basmati rice are grown in India.

← In 1997 American company got patent rights on Basmati Rice through the US Patent and Trademark office.

* Indian parliament recently cleared the 2nd amendment of the Indian Patent Bill which include biopiracy issue, patent terms emergency provisions and research and development initiative.

Ecology

Organism & Populations

* Biomes — mean Annual Temperature — Mean Annual Precipitation

Arctic & Alpine Tundra $\Rightarrow \approx -10^{\circ}$ to 0°C $\Rightarrow \approx 10$ to 120 cm

Coniferous forest $\Rightarrow \approx 0^{\circ}$ to 15°C $\Rightarrow \approx 50$ to 250 cm

Grassland $\Rightarrow \approx 0$ to 25°C $\Rightarrow \approx 50$ to 100 cm

Desert $\Rightarrow \approx 5$ to 20°C $\Rightarrow \approx 10$ to 50 cm

Temperate Forest $\Rightarrow \approx 10$ to 20°C $\Rightarrow \approx 70$ to 230 cm

Tropical Forest $\Rightarrow \approx 20$ to 25°C $\Rightarrow \approx 150$ to 430 cm

* Temperature of tropical desert in summer $> 50^{\circ}\text{C}$

* Salt concentration (in parts per thousand) of $\rightarrow$

$\rightarrow$ inland water $\rightarrow < 5$ (‰)
 $\rightarrow$ sea water $\rightarrow 30-35$
 $\rightarrow$ hypersaline lagoon $\rightarrow > 100$

* In deep sea ($> 500\text{ m}$) source of energy is only hydrothermal vent for life.

* 99% of animals & nearly all plants are conformers.
only 1% animals are regulators.

$\rightarrow$ Birds, mammals & some other animals.

← At high altitude ($> 3500\text{ m}$) a person must have experienced altitude sickness. (Nausea, fatigue, heart palpitation)

* Archaeobacteria (thermo acidophiles) flourish in hot springs and deep sea hydrothermal vents where temperatures far exceeds 100°C .

* In deep sea pressure could be >100 atm.

* Exponential growth $\frac{dN}{dt} = rN$ ($r = b - d$)

* Geometric growth $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$ ($\downarrow$ intrinsic rate)

for Norway rat $r = 0.015$

for flour beetle $r = 0.12$

in 1981 for human population $r = 0.0205$
in India

* In the rocky intertidal community of American Pacific Coast the starfish *Pisaster* is a keystone species. When all starfish were removed, 10 species of invertebrate become extinct within a year because of interspecific competition.

* Nearly 25% of all insects are known to be phytophagous.

* The Abingdon tortoise in Galapagos Islands became extinct within a decade (10 years) after goat were introduced on the island.

* MacArthur show 5 closely related species of warbler were able to avoid competition due to behavioural difference in their foraging activities.

* Human liver fluke depends on two intermediate hosts (a snail and a fish) to complete its life cycle. $\Rightarrow$ Total 3 hosts required.

Ecosystem

- * The annual NPP of whole biosphere is **170** billion tons (dry weight) of organic matter.
- * About **70%** of earth surface is covered by ocean but productivity of oceans are only **55** billion tons.
 $\approx 33\%$ of NPP
It means NPP of land = **115** billion tons in a year.
 67%
- * According to one estimate **4×10^{13} kg** (**40 billion tons**) of carbon is fixed in biosphere through photosynthesis annually.
i.e. $\Rightarrow$ in NPP = **170** billion tons of organic matter
40 billion tons is carbon alone.
- * PAR is **less than 50%** of total incident solar radiation.
- * Plants capture only **2-10%** of PAR.
- * Only **10%** of energy is transferred to each trophic level from the lower trophic level.
- * In pyramid of number of grassland only **3** top-carnivores are supported in an ecosystem based on production of nearly **6 millions** plants.
- * Primary producer (plants) convert only **1%** of the energy in the sunlight available to them into NPP.
- * Carbon constitutes **49%** of dry weight of organism.
- * **71%** of total quantity of global carbon is found dissolve in oceans. **reservoir**
- * Atmosphere contain only **1%** of total global carbon.
- * Robert Constanza and his colleagues have put an average price tag of **US \$ 33 trillion** in a year.

of fundamental ecosystem resources. This is nearly twice the value of global gross national product. (GNP which is US\$ 18 trillion)

* Cost of $\left\{ \begin{array}{l} \text{soil formation} \rightarrow 50\% \\ \text{Recreation} \rightarrow < 10\% \\ \text{Nutrient Cycling} \rightarrow < 10\% \\ \text{climate regulation} \rightarrow 6\% \\ \text{Habitat for wild life} \rightarrow 6\% \end{array} \right\}$ of (33 US\$ trillion)

Biodiversity and conservation

- * Number of species of ants $\rightarrow > 20,000$
beetles $\rightarrow 3,00,000$
fishes $\rightarrow 28,000$
orchids $\rightarrow 20,000$
- * India has ^(2 lakh) more than 50,000 genetically different strain of rice and 1000 varieties of mango.
- * According to IUCN (2004) total number of plants and animal species described so far is more than 1.5 million.
- * According to Robert May's estimate the global species diversity is about 7 million. (more conservative & scientific)
- * According to some extreme estimate total number of species on earth range from 20 to 50 millions.
- * In total discovered species more than 70% ^{$\rightarrow 72\%$} are animals. And plants are not more than 22%.
- * Among animals insects are 70% of the total. i.e. out of every 10 animals on this planet 7 are insects.
- * India $\left\{ \begin{array}{l} \rightarrow 2.4\% \text{ of world's land area.} \\ \rightarrow 8.1\% \text{ of global species diversity.} \end{array} \right.$

- * India is one of the 12 mega diversity country of world.
- * From India → 45,000 species of plants and twice (90,000) species of animals are recorded.

* According to Robert May's estimation, only 22% of total species have been recorded so far.

→ In India → more than 1 lakh plant species and more than 3 lakh animal species yet to be discovered.

* Scientists estimate that in rain forests (Tropical rain forests) there might be at least 2 million insect species waiting to be discovered and named.

- * Tropical region (23.5° N to 23.5° S) → maximum No. of species.
- Colombia (located near equator) → 1400 species of birds.
- New York (41° N) → 105 species of birds.
- Greenland (71° N) → 56 species of birds.
- India (Temperate & Tropical region) → > 1200 species of birds.
(less) (more)

* Tropical forest (like Ecuador) has upto 10 times as many species of angiosperms as forest of equal area in temperate region (like Midwest of USA).

* Tropical Amazonian rain forest of South America

→ Plants species	> 40,000
→ Fishes "	> 3,000
→ Birds "	> 1300
→ Mammals "	> 427
→ Amphibians "	> 427
→ Reptiles "	> 378
→ Invertebrates "	> 1,25,000

* Alexander von Humboldt gives species area relation -

$$S = CA^Z$$

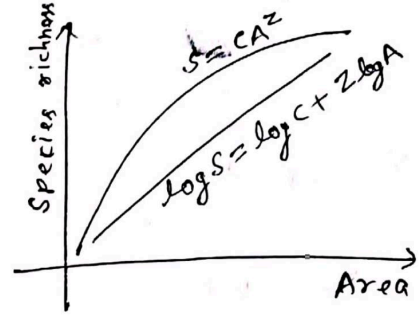
$$\log S = \log C + Z \log A$$

S = species richness

A = area

Z = slope (regression coefficient)

C = intercept (Y-axis)



* value of Z lies in the range of **0.1 to 0.2**.

* for large area $Z \Rightarrow$ **0.6 to 1.2**

* For fruit eating birds and mammals in tropical forests Z is found to be **1.15**.

* The colonisation of tropical Pacific Islands by Humans is said to have led to the extinction of **more than 2000** native birds' species.

* The IUCN Red list (2004) documents the extinction of **784** species in last **500** years.

→ **338** vertebrates
 → **359** invertebrates } Animals species (**697**)
 → **87** plant species

* **27** species disappear within last **20** years alone.

* **More than 15,500** species world wide are facing the threat of extinction.

↓
(of which **more than 650** are from India)

includes → **12%** of birds species
 → **23%** of all mammals species,
 → **32%** of all amphibian species.
 → **31%** of all gymnosperm species.

* Since the origination & diversification of life on earth there were **5** episodes of mass extinction of species.
 → revolutions

* Current species extinction rates are estimated to be **100 to 1000** times faster than pre human times.
 → Half of all sps on earth might be wiped out within 100 years.

* Tropical rain forests once covering more than 14% of earth's land surface but now it cover no more than 6%. $\Rightarrow$ due to habitat loss & fragmentation.

* The Nile perch introduced into Lake Victoria (in east Africa) led more than 200 species of cichlid fish become extinct.

* More than 25% of the drugs currently sold in the market world-wide are derived from plants and 25,000 species of plants contribute to the traditional medicines used by native peoples around the world.

* Amazon forest $\rightarrow$ produce 20% of total oxygen (Lungs of earth) on earth by photosynthesis.

* Biodiversity Hotspot $\rightarrow$ maximum species diversity
 $\downarrow$
Covers less than 2% of earth's land area,
 $\rightarrow$ initially 25 were identified
 $\rightarrow$ later 9 more added
 $\rightarrow$ now total 34 in the world
 $\rightarrow$ 3 in India
 $\rightarrow$ Western Ghats-Sri Lanka
 $\rightarrow$ Indo-Burma
 $\rightarrow$ Himalaya

* Strict protection of hotspot reduce mass extinction of sps by 30%.

* Earth Summit (1992) $\rightarrow$ for biodiversity conservation & sustainable utilization of its benefits.
 $\hookrightarrow$ Rio de Janeiro

* World Summit (2002) $\rightarrow$ for sustainable development of biodiversity
 $\hookrightarrow$ Johannesburg South Africa

$\Downarrow$
190 countries pledged their commitment to achieve by 2010.

In India

- Biosphere Reserve $\Rightarrow > 14$
- National Parks $\Rightarrow > 90$
- Wildlife Sanctuaries $\Rightarrow > 450$

Environmental Issues

- * 1972 $\Rightarrow$ National Committee for Environmental planning & Coordination
- * 1974 $\Rightarrow$ Water (prevention and control of pollution) Act.
- * 1974 $\Rightarrow$ Chipko movement
- * 1980s $\Rightarrow$ Joint Forest Management
- * 1981 $\Rightarrow$ Air (prevention & control of pollution) act.
- * 1984 $\Rightarrow$ Bhopal Tragedy
- * 1984 $\Rightarrow$ Ministry of Environment & Forest
- * 1986 $\Rightarrow$ Environment (protection) Act (NEPA).
- * 1987 $\Rightarrow$ Noise included in air pollution.
- * 1987 $\Rightarrow$ Montreal protocol (Canada) $\Rightarrow$ ozone depletion
- * 1988 $\Rightarrow$ National forest policy
- * 1989 $\Rightarrow$ Montreal protocol become effective.
- * 1997 $\Rightarrow$ Kyoto protocol for climate change (COP-3)

* Estimated maximum concentration (in ppm) of metals in drinking water $\rightarrow$

Cd	$\rightarrow 0.005$
Mn	$\rightarrow 0.05$
Fe	$\rightarrow 0.2$
Al	$\rightarrow 0.2$
Cu	$\rightarrow 3.0$
Zn	$\rightarrow 5.0$

(1 ppm = 1000 ppb)

- * El electrostatic precipitator can remove 99% particulate matter in exhaust from a thermal power plant.
- * According to CPCB particulate size 2.5 μ m (or) PM_{2.5} less is harmful for human.

* Exposure to extremely high sound level, 150 dB @ more generated by take off of a jet plane @ rocket may damage ear drum.

* In 1990s Delhi ranked 4th among the 41 most polluted cities of the world.

* All buses of Delhi were converted to run off CNG by the end of 2002. (COP-8 held in Delhi)

* Euro II Norms $\begin{cases} \rightarrow \text{Sulphur in petrol} \Rightarrow 150 \text{ ppm} \\ \rightarrow \text{Sulphur in diesel} \Rightarrow 350 \text{ ppm} \end{cases}$
(1 April 2005 से)

* Aromatic hydrocarbons are to be contained at 42% of the concerned fuel.

* The goal, according to the roadmap, is to reduce sulphur to 50 ppm in petrol & diesel and bring down the level to 35% of aromatic hydrocarbons.

* From 1 April 2005 $\begin{cases} \rightarrow \text{Euro II} \Rightarrow \text{All over India} \\ \rightarrow \text{Euro III} \rightarrow 11 \text{ cities.} \end{cases}$

* From 1 April 2010 $\begin{cases} \rightarrow \text{Euro III} \rightarrow \text{All over India} \\ \rightarrow \text{Euro IV} \rightarrow 11 \text{ cities} \end{cases}$

Delhi, Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Ahmedabad, Pune, Surat, Kanpur, Aggra

* According to an estimate, a substantial fall in CO_2 and SO_2 level has been found in Delhi between 1997 and 2005.

: A mere 0.1% impurities make domestic sewage unfit for human use.
water

- * DO in cold water $\rightarrow$ 10 ppm
- * if D.O. of water is below 6 ppm growth of fishes get inhibited.
- * BOD of clean water is less than 5 ppm.
- * Highly polluted water could have BOD value 17 ppm or more.

* Integrated Waste Water Treatment

$\hookrightarrow$ In Arcata (North coast of California)

2 Stage

- $\hookrightarrow$ Sedimentation and filtration of pollutants.
- $\hookrightarrow$ Neutralisation, Absorption and assimilation of pollutants (heavy metal) by 6 connected marshes over 60 hectares of marshland.
 - $\hookrightarrow$ Plants, algae, Fungi & Bacteria are grown.

* Case study of Remedy for plastic waste $\Rightarrow$ Bangalore

* Ahmed Khan (Plastic sac manufacturer)

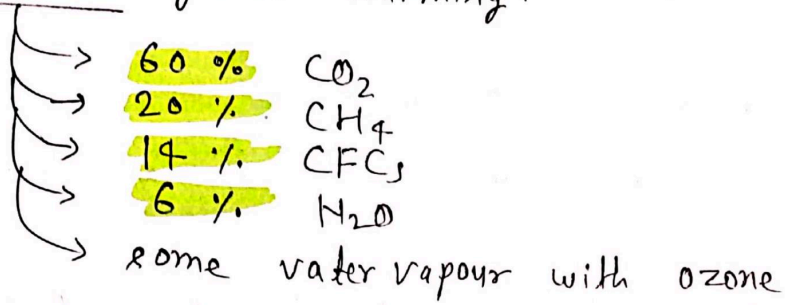
- $\hookrightarrow$ Polyblend $\rightarrow$ fine powder of recycled modified plastic
 - $\hookrightarrow$ enhance the water repellent property of Bitumen. And increase road life by a factor of 3.

* Using Khan's technique, by year 2002, more than 40 kms road in Bangalore has already been laid.

* It has been recommended that storage of nuclear waste after sufficient pre-treatment should be done in suitably shielded containers buried within the rocks, about 500m deep below earth's surface.

* Without greenhouse effect the average temperature at surface of Earth would have been a chilly -18°C rather than the present average of 15°C .

* Relative contribution of various greenhouse gases to total global warming.



* During the past century the temperature of Earth has increased by 0.6°C , most of it during ~~last 30 years~~ last 30 years.

* According to an estimate, almost 40% forests have been lost in the tropics, compared to only 1% in the temperate region.

* In India — at beginning of 20th century $\Rightarrow$ forest covers 30% land area.
By the end of 20th century $\Rightarrow$ forest covers only 19.4% land area.

* National forest policy (1988) of India has recommended 33% forest cover for plains and 67% for the hills.

* In 1731 many people of Bishnoi community sacrificed their lives saving trees. (Jodhpur)

NCERT Biology

→ Yearwise data

1731 ⇒ Many people of Bishnoi community sacrifices their lives ⇒ saving tree (Jodhpur)

1770 ⇒ Priestley's experiment ⇒ Bell Jar, Mice, mint plant, Candle.

1774 ⇒ Discovery of oxygen by Joseph Priestley.

1838 ⇒ Matthias Schlieden } Gives Cell Theory
1839 ⇒ Theodor Schwann }

1850s ⇒ Before industrialisation ⇒ more white wing moth

1854 ⇒ Julius von Sachs proved that glucose is formed in photosynthesis & stored as starch.

1855 ⇒ "Omnis cellula e cellula" by Rudolf Virchow & complete the cell theory.

1856 - 1863 ⇒ Mendel work on garden pea.

1860 ⇒ Hydroponics technique developed by Julius von Sachs.

1865 ⇒ Mendel publishes his work on inheritance of characters.

1866 ⇒ Down's Syndrome discovered by Langdon Down.

1869 ⇒ DNA was 1st identified by Friedrich Miescher & termed as "Nuclein".

1891 ⇒ Henking discovered X-body (X-chromosome) in insect. (XO-type sex determination)

1892 $\Rightarrow$ Discovery of TMV by D.J. Ivanowsky.

1898 $\Rightarrow$ M.W. Beijerinck demonstrate infectious nature of TMV

$\rightarrow$ Discovery of Golgi Body by Camillo Golgi.

1900 $\Rightarrow$ Rediscovery of Mendel's work by three scientists

$\rightarrow$ Hugo de Vries
 $\rightarrow$ Karl Correns
 $\rightarrow$ Tschermak

1902 $\Rightarrow$ Chromosomal theory of inheritance given by Sutton & Boveri.

1905 $\Rightarrow$ Law of limiting factor for plants given by Blackmann.

1920 $\Rightarrow$ After Industrialisation $\Rightarrow$ more black moth

1928 $\Rightarrow$ Fredrich Griffith's Transformation experiment on *Streptococcus pneumoniae* bacteria.

1933-1944 $\Rightarrow$ Avery, Macleod & McCarty use Transformation principle of Griffith's experiment & conclude that DNA is Genetic material.

1938 $\Rightarrow$ Coelacanth/Lobefinns Caught in South Africa
 $\rightarrow$ Ancestors of amphibia.

1947 $\Rightarrow$ At the time of our independence India's population was approximately 350 million.

- 1950s $\Rightarrow$ Electron microscope invention
 $\Rightarrow$ Tissue culture started
- 1951 $\Rightarrow$ Family planning were initiated
- 1952 $\Rightarrow$ Hershey & Chase experiment more correctly prove that DNA is genetic material.
- 1953 $\Rightarrow$ Discovery of ribosome by George Palade
 $\Rightarrow$ S.L. Miller Experiment of evolution.
 $\Rightarrow$ Double helical structure model of DNA given by Watson & Crick.
- 1958 $\Rightarrow$ Matthew Meselson & Franklin Stahl experiment (E. coli in NH_4Cl medium) prove that DNA replication is semiconservative
 $\Rightarrow$ Taylor study on vicia faba plant $\Rightarrow$ use radioactive thymidine and prove semiconservative nature of DNA replication.
- 1960s $\Rightarrow$ Green Revolution.
- 1963 $\Rightarrow$ Restriction endonuclease isolated from E. coli.
 $\Rightarrow$ Wheat varieties Sonalika & Kalyan Sona (High yield and Diseases resistant) introduced in India.
- 1964 $\Rightarrow$ 1st time NCERT Biology Book for senior secondary was published.
- 1966 $\Rightarrow$ Derived semidwarf varieties of rice introduced in India.
- 1971 $\Rightarrow$ Discovery of viroids by T. O. Diener
 $\Rightarrow$ Govt. of India legalised MTP.
- 1972 $\Rightarrow$ Fluid Mosaic model of plasma membrane by Singer & Nicolson.
 $\Rightarrow$ 1st recombinant DNA developed by Stanley Cohen and Herbert Boyer.
 $\Rightarrow$ National committee for environmental planning and coordination was established.
 $\Rightarrow$ Wild life protection act in India.

1974 $\Rightarrow$ Water Act.
 $\Rightarrow$ Chipko movement

1980s $\Rightarrow$ Joint Forest Management

1981 $\Rightarrow$ Air act.
 $\Rightarrow$ AIDS was first reported
 $\Rightarrow$ Intrinsic growth rate of India $r = 0.0205$

1983 $\Rightarrow$ Eli Lilly an American company prepare human Insulin from E. coli.

1984 $\Rightarrow$ Bhopal Tragedy
 $\Rightarrow$ Ministry of Environment and Forest established

1986 $\Rightarrow$ Environment Act (NEPA)

1987 $\Rightarrow$ Noise included in air pollution.
 $\Rightarrow$ Montreal Protocol (Canada)

1988 $\Rightarrow$ National Forest Policy

1989 $\Rightarrow$ Montreal Protocol become effective.

1990s $\Rightarrow$ Delhi ranked 4th among 41 most polluted cities.

1990 → First clinical gene therapy was given to a 4 year old girl with ADA deficiency.
→ Human Genome Project (HGP) launched.
→ World's population was 2 billion.

1992 → Earth Summit for Biodiversity conservation in Rio de Janeiro.

1997 → Kyoto Protocol on climate change (COP-3)
→ An American company got patent rights on Basmati Rice through the US patent and Trademark office.
→ First Transgenic cow "Rosie" produced human milk protein.

2000 → world's population 6 billion & India's population 2 billion.
→ Using Ahmad Khan's technique >40 km road laid.

2002 → World Summit for sustainable development of biodiversity (Johannesburg South Africa)
→ All Buses of Delhi were converted to run of CNG by the end of the year. (COP-8)

2003 → HGP completed

2004 → According to IUCN → 1.7 to 1.8 million species on earth
→ Extinction of 789 species in last 500 years.

2005 → 1 April → Euro-II → all over INDIA.
→ Euro-III → 13 most polluted city of INDIA.

2006 → Sequencing of 1st chromosome number completed in May (HGP)

2010 $\Rightarrow$ 1 April $\begin{cases} \rightarrow \text{Euro-III} \rightarrow \text{all over India} \\ \rightarrow \text{Euro-IV} \rightarrow 13 \text{ cities} \end{cases}$

2011 $\Rightarrow$ world's population - 7.2 billion
India's population - 1.2 billion

2017 $\Rightarrow$ MTP (Amendment) Act.

Thank You